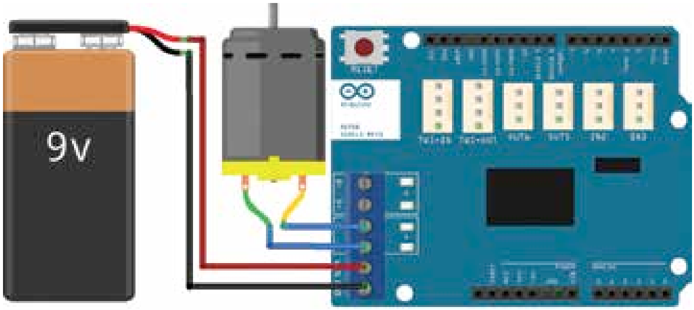

Connecting a DC motor to a Motor Shield

stepping motors as well. It needs a separate power source to supply power to the connected motors. The shield can supply maximum 2 A of current per motor. A total of 4 A current for two motors can be connected.

The shield has two separate channels (called A and B) and each uses four of the Arduino™ pins to drive or sense the motor. In total, there are eight pins in use on this shield. You can use each channel separately to drive two DC motors or combine them to drive one bipolar stepper motor.

The shield's pins, divided by channel, are shown in the table below:

Operating a motor:

- To use the motor, the brake input of the shield should be LOW to allow operation of the motor.
- Speed control of the connected motor can be achieved by generating a PWM wave of the required average voltage to attain required speed of the motor.
- The direction of rotation of the motor is decided by the input on the DIRECTION pin
    - A HIGH input is used for forward rotation
    - A LOW input is used for backward rotation

FUNCTION	CHANNEL A	CHANNEL B
Direction	Digital 12	Digital 13
Speed (PWM)	Digital 3	Digital 11
Brake	Digital 9	Digital 8
Current Sensing	Analog 0	Analog 1

Code for driving a DC motor

If you don't need the brake and the current sensing and you also need more pins for your application, you can disable this features by cutting the respective jumpers on the back of the shield.